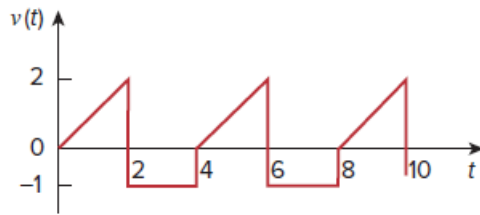


Problem

Compute the rms value of the waveform depicted in Fig. 11.61.

Figure 11.61:



Step-by-step solution

Step 1 of 1

$$V(t) = \begin{cases} t & 0 < t < 2 \\ -1 & 2 < t < 4 \end{cases}$$

$$V_{\text{rms}}^2 = \frac{1}{4} \left[\int_0^2 t^2 dt + \int_2^4 (-1)^2 dt \right]$$

$$V_{\text{rms}}^2 = \frac{1}{4} \left[\frac{8}{3} + 2 \right]$$

$$V_{\text{rms}}^2 = 1.1667$$

$$\therefore \boxed{V_{\text{rms}} = 1.08 \text{ V}}$$